

Clinical Diagnosis & History:

Invasive duct carcinoma left breast 5:0 and severe ADH left breast 11:00 (on core biopsy); for left total mastectomy, SLNB, possible left ALND and immediate reconstruction.

Specimens Submitted:

- 1: SENTINEL NODE #1, LEVEL ONE LEFT AXILLA, EXCISION:
- 2: SENTINEL NODE #2, LEVEL ONE LEFT AXILLA, EXCISION
- 3: NON-SENTINEL NODE, LEFT AXILLA, EXCISION
- 4: BREAST, LEFT, SIMPLE MASTECTOMY

UUID:4022B2EB-04C3-43F6-9D2C-1A47D80EF6EE
TCGA-A0-A037-01A-PR

Redacted



DIAGNOSIS:

SUMMARY OF FINDINGS IN THIS CASE:

Invasive ductal carcinoma present as two foci spanning 28 mm and 2 mm and is 3.6 cm from the closest inked margin.

The nipple dermis is involved by invasive ductal carcinoma.

Mucinous carcinoma spans 2.5 mm and does not involve the inked margin.

All foci of invasive carcinoma (ductal and mucinous) are ER positive, HER2 negative.

DCIS is present and is 3.6 cm from the closest inked margin.

Sentinel lymph nodes: 2/3.

Non-sentinel lymph nodes: 1/1.

Total lymph nodes: 3/4.

Type of metastasis: macrometastasis (6 mm) to an intramammary lymph node

1. SENTINEL NODE #1, LEVEL ONE LEFT AXILLA, EXCISION:

Sentinel Lymph Node Status:

Number of Sentinel Lymph Nodes Examined: 1

Number of Sentinel Lymph Nodes with carcinoma: 1

Size of Largest Metastatic Focus:

>= 0.2 mm to < 2 mm (micrometastasis)

The metastatic focus spans 0.4 mm

Level of Detection:

Initial H&E stained section

Extracapsular Extension:

Not present

2. SENTINEL NODE #2, LEVEL ONE LEFT AXILLA, EXCISION:

** Continued on next page **

1CD-0-3

carcinoma, infiltrating duct and mucinous

8523/3

Site: breast, NOS CSD.9 fw
10/22/11

Criteria	Yes	No
Diagnosis Discrepancy		☒
Primary Tumor Site Discrepancy		☒
IPAA Discrepancy		☒
Prior Malignancy History		☒
Local/Synchronous Primary Nodule		☒
Case Is (circle):	QUALIFIED	DISQUALIFIED
Reviewer Initials	10/22/11	

Sentinel Lymph Node Status:

Number of Sentinel Lymph Nodes Examined: 1

Number of Sentinel Lymph Nodes with carcinoma: 1

Size of Largest Metastatic Focus:

> 2 mm

The metastatic focus spans 3 mm

Level of Detection:

Initial H&E stained section

Extracapsular Extension:

Present, measuring < 2mm in largest diameter

The metastatic carcinoma is negative for WT1.

3. Non-sentinel node, left axilla, excision:
- One benign lymph node (0/1).

4. BREAST, LEFT, SIMPLE MASTECTOMY:

Invasive Carcinoma:

Ductal, NOS type

Focal mucinous carcinoma (see note)

Histologic Grade:

III/III: Minimal or no tubule formation (< 10% of tumor)

Nuclear Grade:

III/III (marked variation in size and shape)

Tumor Size:

Two foci (size specified below)

the invasive ductal carcinoma spans 28 mm and 2 mm

Ductal carcinoma in situ (DCIS):

Present

DCIS, Architecture:

Solid

Papillary

DCIS Nuclear Grade:

High

Necrosis in DCIS:

Minimal

Extensive intraductal component (>25% of tumor mass):

Not Identified

Lobular Neoplasia:

Lobular carcinoma in situ (LCIS), classical type

Location of Invasive Carcinoma:

Upper inner quadrant (UIQ)

Lower inner quadrant (LIQ)

Location of DCIS:

Upper inner quadrant (UIQ)

Lower inner quadrant (LIQ)

Nipple Involvement:

Invasive carcinoma involves the nipple stroma

Skin:

Tumor emboli are present in dermal lymphatic channels

The examined skin margins are uninvolved by carcinoma

Calcification:

In benign breast tissue

In invasive carcinoma

Lymphovascular Invasion:

** Continued on next page **

Present

Surgical Margins:

Invasive carcinoma is 36 mm from the closest margin
DCIS is 36 mm from the closest margin

Benign Breast Tissue:

Atypical ductal hyperplasia (ADH)
Fibrocystic changes, including apocrine metaplasia, cyst formation
and stromal fibrosis
Biopsy site changes
Columnar cell changes

Lymph Nodes:

LN - Number of lymph nodes with metastatic carcinoma: 1
Number of lymph nodes examined: 1

The mass identified in the UOQ at 1:00 is an intramammary lymph node, almost entirely replaced by metastatic carcinoma. The metastatic carcinoma measures 0.6 cm and is negative for WT1. Results of immunostains for calponin and p63 support the diagnosis.

A 2.5 mm focus of invasive mucinous carcinoma is also identified, with nuclear grade II/III.

Immunohistochemical stains were performed on formalin-fixed tissue with the following results for

INVASIVE DUCTAL CARCINOMA (block 4-5):

ESTROGEN RECEPTOR (6F11, 90% nuclear staining with strong intensity

PROGESTERONE RECEPTOR (1E2, negative (no nuclear staining)

HER2 (4B5, Negative (0 / 1+)

The invasive ductal carcinoma is negative for WT1.

The smaller focus of invasive ductal carcinoma has similar immunoprofile. Positive membranous immunoreactivity for E-cadherin supports ductal differentiation.

MUCINOUS CARCINOMA (block 4-19):

ESTROGEN RECEPTOR (): 95% nuclear staining with strong intensity

PROGESTERONE RECEPTOR (95% nuclear staining with strong intensity

HER2 (Negative (0)

The mucinous carcinoma is positive for WT1.

Comment: Controls are satisfactory. PATHWAY anti-HER-2/neu is an FDA-approved rabbit monoclonal primary antibody (clone 4B5) directed against the internal domain of the c-erbB-2 oncoprotein (HER2) for immunohistochemical detection of HER2 protein overexpression in breast cancer tissue routinely processed for histologic evaluation. The HER2 test results are reported in accordance with the ASCO/CAP guideline recommendations for HER2 testing in breast cancer (J Clin Oncol 2007; 25(1):118-145). The ER and PR rabbit monoclonal antibodies are also FDA approved.

** Continued on next page **

I ATTEST THAT THE ABOVE DIAGNOSIS IS BASED UPON MY PERSONAL EXAMINATION OF THE SLIDES (AND/OR OTHER MATERIAL), AND THAT I HAVE REVIEWED AND APPROVED THIS REPORT.

*** Report Electronically Signed Out ***

Gross Description:

1) The specimen is received fresh for frozen section consultation, labeled "sentinel node #1, level one, left axilla" and consists of a single node measuring 1 cm. Bisected and entirely submitted for frozen section.

Summary of sections:
FSC -- frozen section control

2) The specimen is received fresh for frozen section consultation, labeled "sentinel node #2, level one, left axilla" and consists of a single node measuring 2.0 cm. Bisected and entirely submitted for frozen section.

Summary of sections:
FSC -- frozen section control

3) The specimen is received fresh for frozen section consultation, labeled "non-sentinel node left axilla" and consists of a single fatty node measuring 0.7 cm. Bisected and entirely submitted for frozen section.

Summary of sections:
FSC -- frozen section control

4.) The specimen is received fresh labeled, "Left total mastectomy, suture marks axillary aspect" and consists of a previously incised, 24.1 x 19.5 x 4.6 cm breast with a partially overlying, 11.7 x 3.5 x 0.1 cm skin ellipse. Situated centrally on the skin surface is a 1.2 x 1.2 x 0.1 cm, crusty, partially inverted, diffusely indurated, nipple with smooth, pink-white suspicious cut surfaces surrounded by a 4.7 x 3.4 cm areola. The wrinkled, tan-white skin is free of any gross abnormalities. A suture demarcates the 7.2 x 4.6 x 0.5 cm axillary aspect. The posterior surface of the breast is inked black, the anterior blue and the axillary aspect is inked yellow. The specimen is serially sectioned to reveal a 2.8 x 2.4 x 2.1 cm, firm, lobular to fatty infiltrating bordered, pink-white mass, located in the lower inner and outer quadrants, from 5 o'clock to 7 o'clock, underlying the nipple, 0.5 cm from the nipple base and 3.6 cm from the deep margin, (designated Mass #1). The mass grossly appears to be separate from the suspicious indurated nipple. Also, 5.9 cm superior and 1.5 cm medial to Mass #1, is a 2.5 x 2.2

** Continued on next page **

x 1.3 cm, biopsy site partially surrounded by moderately dense, focally; cystic, nodular and hemorrhagic, white fibrous tissue, located within the upper outer quadrant, at 1 o'clock, 5.1 cm the nipple and 3.2 cm from the deep margin, (designated Biopsy site #1). In addition, partially abutting biopsy site #1 is a twist-clip situated within a 2.6 x 1.6 x 1.4 cm well-healed biopsy site with a peripherally attached 0.9 x 0.9 x 0.7 cm, ovoid, firm, smooth to nodular bordered, white to indigo blue ink-tinged mass, located within the upper outer quadrant, at 2 o'clock, 5.5 cm from the nipple and 4.2 cm the deep margin (designated Biopsy site #2 and Mass #2). The remaining breast parenchyma consists of lobules of yellow adipose separated by bands of mild to moderately dense, focally nodular and cystic, white fibrous tissue. Sectioning of the axillary aspect reveals no grossly identifiable lymph nodes. TFS is submitted of Mass #1, per protocol Representative sections are submitted. Additional sections are submitted following microscopic review of the specimen.

Summary of sections:

N - nipple
NB - nipple base
D - deep margin
M1 Mass #1
BX1 Biopsy site #1
BX12 Biopsy sites #1 & #2
BX2 -- Biopsy site #2
M2BX2 Mass #2 with Biopsy site #2
M2 Mass #2
UIQ - upper inner quadrant
LIQ - lower inner quadrant
UOQ - upper outer quadrant
LOQ - lower outer quadrant
AD12 skin additional sections, 12 o'clock
AD3 skin additional sections, 3 o'clock
AD6 skin additional sections, 6 o'clock
AD9 skin additional sections, 9 o'clock

Summary of Sections:

Part 1: SENTINEL NODE #1, LEVEL ONE LEFT AXILLA, EXCISION:

Block	Sect.	Site	PCs
1		FSC	1

Part 2: SENTINEL NODE #2, LEVEL ONE LEFT AXILLA, EXCISION

Block	Sect.	Site	PCs
1		FSC	1

Part 3: NON-SENTINEL NODE, LEFT AXILLA, EXCISION

Block	Sect.	Site	PCs
1		FSC	1

** Continued on next page **

Part 4: BREAST, LEFT, SIMPLE MASTECTOMY

Block	Sect. Site	PCs
1	AD12	1
1	AD3	1
1	AD6	1
1	AD9	1
4	BX1	5
1	BX12	1
1	BX2	1
1	D	1
1	LIQ	2
1	LOQ	2
6	M1	6
1	M2	1
1	M2BX2	1
2	N	5
1	NB	1
1	UIQ	2
1	UOQ	2

Intraoperative Consultation:

Note: The diagnoses given in this section pertain only to the tissue sample examined at the time of the intraoperative consultation.

1. FROZEN SECTION DIAGNOSIS: SP: Sentinel node #1 level 1 left axilla (fs): Benign

PERMANENT DIAGNOSIS: SEE FINAL DIAGNOSIS

2. FROZEN SECTION DIAGNOSIS: SP: Sentinel node #2 level 1 left axilla (fs): Benign

PERMANENT DIAGNOSIS: SEE FINAL DIAGNOSIS

3. FROZEN SECTION DIAGNOSIS: SP: Non-sentinel node left axilla (fs): Benign

PERMANENT DIAGNOSIS: SAME

** End of Report **